SCIENCE 1.3 **Name:** ***ANSWERS***

WORKSHEET SIX **INHERITANCE**

1. Match up the following terms with their appropriate descriptors.

(a) Genotype	**2**	(1) A gene that carry identical alleles.
(b) Zygote	**7**	(2) The pattern of genes present in cells
(c) Phenotype	**8**	(3) An allele whose influence is more powerful than its partner
(d) Recessive allele	**6**	(4) A pair of alleles that carry different forms of the same gene
(e) Homozygous	**1**	(5) A sex cell
(f) Heterozygous	**4**	(6) An allele that is expressed only if the dominant form is absent
(g) Gamete	**5**	(7) First cell of the new organism
(h) Dominant allele	**3**	(8) The actual appearance produced by an organism's genes

2. Complete the paragraph using the words in the key list below.

shape, variation, body height, symmetrical, population, normal, bell.

Many characteristics are found to display **continuous** ***variation***. Examples would be body mass and ***body height***. When a continuous variation pattern is graphed, the graph always displays the same ***shape***. This is called a ***normal*** distribution curve or a ***bell*** curve. However if the data for the graph is taken from too small a ***population*** the graph may not have a ***symmetrical*** shape.

3. Read the following genetic "rules" for monohybrid crosses. Beside each rule put a **T** or a **F** depending on whether you think it is true or false.

(a) Traits are controlled by a pair of genes. ***T***

(b) Recessive alleles mask dominant ones. ***F***

(c) Gametes have one allele for each trait. ***T***

(d) Different forms of a gene for the same trait are called alleles. ***T***

(e) The genotype describes the appearance. ***F***

(f) The phenotype lists the alleles present. ***F***

(g) In heterozygous individuals one allele is dominant and one is recessive. ***T***

(h) An individual displaying the dominant trait could have one of two genotypes. ***T***

4. In guinea-pigs, a black colour coat (B) is dominant to a brown colour coat (b).

(a) Write the genotype of a homozygous dominant individual. ***BB***

(b) Write the genotype of a heterozygous individual ***Bb***

(c) Write the genotype of a homozygous recessive individual ***bb***

(d) A heterozygous black male was mated with a homozygous brown female. Complete the punnet square to show the possible offspring genotypes.

	B	***b***
b	***Bb***	***bb***
b	***Bb***	***bb***

(e) What ratio of brown to black offspring is suggested by the the completed punnet square. ***1:1***

5. Humans either have the ability to roll their tongue (T) or they do not (t). Tongue rolling is dominant over non-tongue rolling. A married couple who are both tongue rollers have three children. One child is a non-tongue roller and the other two are tongue rollers.

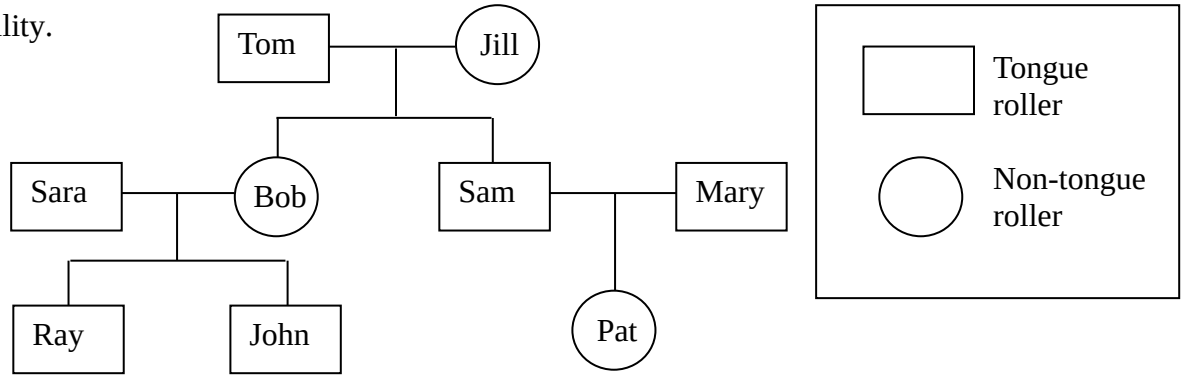
(a) Draw a punnet square that shows how this could happen.

	T	***t***
T	***TT***	***Tt***
t	***Tt***	***tt***

(b) Is it possible for one of the children to be heterozygous? ***Yes*** . Write this child's genotype. ***Tt***.

(c) What is the phenotype ratio derived from this punnet square? ***Tongue roller : non-tongue roller = 3:1***

6. An inheritance chart is shown below for three generations of a family regarding tongue-rolling ability.



(a) Write genotypes for Tom ***Tt*** and for Jill ***tt***.

(b) How did you know whether Tom was Tt or TT? ***Bob is Tom's son and he has a genotype tt so he must have got t from Tom***

(c) Write genotypes for Sam ***Tt*** and Mary ***Tt*** and John ***Tt***.

(b) How did you know whether Mary was Tt or TT? ***Pat is Mary's daughter and she has a genotype tt so she must have got t from Mary***